

Advanced ROMs in SciML

SISSA PhD Course

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9 June 2026



Schedule

Day	Time	Type	Room	Topic
Tue 9 June 2026	11-13	Lecture	134	Introduction to Scientific Machine Learning
Tue 9 June 2026	14-16	Lecture	134	<i>FFN, CNN, backpropagation, optimization</i>
Thu 11 June 2026	11-13	Lecture	134	From projection based to data-driven ROMs
Thu 11 June 2026	14-16	Lecture	134	<i>RBniCS to MLniCS</i>
Fri 19 June 2026	11-13	Lecture	136	GNNs based architectures 1
Fri 19 June 2026	14-16	Lecture	005	<i>GCA-ROM, GFN, MeshGraphNet</i>
Tue 23 June 2026	11-13	Lecture	134	GNNs based architectures 2
Tue 23 June 2026	14-16	Lecture	134	<i>GCA-ROM, GFN, MeshGraphNet</i>
Thu 25 June 2026	11-13	Lecture	134	Time-dependent problems and Neural Operators
Thu 25 June 2026	14-16	Lecture	134	<i>Sindy, NODEs, DMD, OpInf, DeepONet, FNO</i>

Exam

- Completely open to your interests and needs!
- Develop a useful and challenging parametric benchmark
- Compare different dimensionality reduction strategies
- Try to reproduce the results from a paper and investigate different methodologies
- Send a complete and commented report of the .ipynb project
- No deadlines on my side

References

1. Quarteroni, A., Gervasio, P., Regazzoni, F., 2025. Combining physics-based and data-driven models: advancing the frontiers of research with scientific machine learning. Math. Models Methods Appl. Sci. <https://doi.org/10.1142/S0218202525500125>
2. Prince, S. Understanding Deep Learning, 2025. MIT Press. <https://mitpress.mit.edu/9780262048644/understanding-deep-learning/>
3. Bach, F. Learning Theory from First Principles, 2024. MIT Press. <https://mitpress.mit.edu/9780262049443/learning-theory-from-first-principles/>
4. Peyré, G., 2022. Mathematical foundations of data sciences. <https://www.numerical-tours.com>
5. Nielsen, M.A., 2015. Neural Networks and Deep Learning. <http://neuralnetworksanddeeplearning.com>